200 INCHES THE DIAMETER OF THE PRIMARY MIRROR OF THE HALE TELESCOPE

For 45 years, the Hale telescope was the world's biggest effective telescope. It is still in use today, collecting data on around 290 nights every year.

Proton linear accelerator
The first device to accelerate protons in a straight line was a 40 ft (12 m) construction. It helped advance research and understanding of fundamental particles of matter.

theoretical technique for producing 3D-style images, or holograms. He described how to produce an image of an object that changed orientation depending on the viewing angle. However, practical application of his theory became possible only with the development of the first working lasers in 1960.

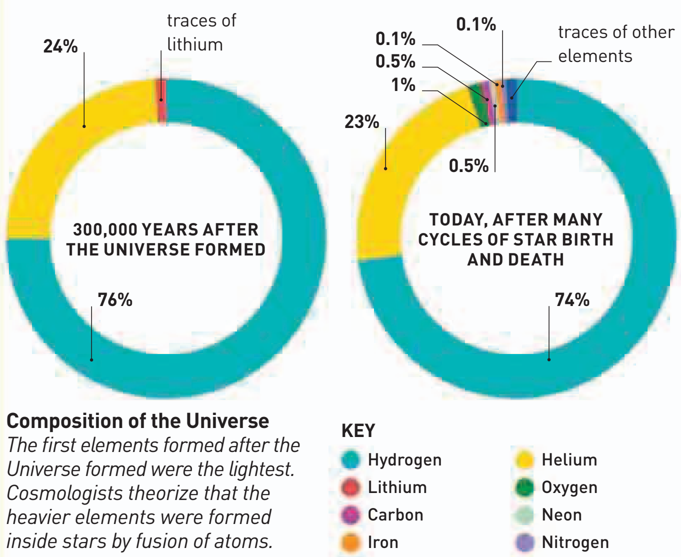
American engineer **Percy Spencer** (1894–1970) was involved in **radar design** with a company called Raytheon. He accidentally discovered that the microwaves produced by a vacuum tube called a magnetron—a core component of radar—heated food. By confining the food "target" in a metal box, Spencer invented the **first microwave oven**—patented in 1945. Raytheon sold the first commercial model in 1947.

IN MARCH, American physicists **Julian Schwinger** (1918–94) and **Richard Feynman** (1918–88) introduced a new field of science: **quantum electrodynamics**. It described how electrically charged particles interacted with packets of electromagnetic radiation, called photons. The conference included the first presentations of **"Feynman diagrams"** to show interactions between subatomic particles, with time along one axis and space along another.

In April, Russian physicist **George Gamow** (1904–68) and American cosmologist **Ralph Alpher** (1921–2007) proposed that elements produced when the Universe formed (see pp.344–45) occurred in fixed proportions. Hydrogen and helium are still the most abundant elements in the Universe today.

In June, Caltech's (California Institute of Technology) Palomar Observatory was completed. Its dome houses the **Hale telescope**, named after American **astronomer George Hale**. This telescope—first used by astronomer Edwin Hubble—has aided in the discovery of quasars and stars of distant galaxies.

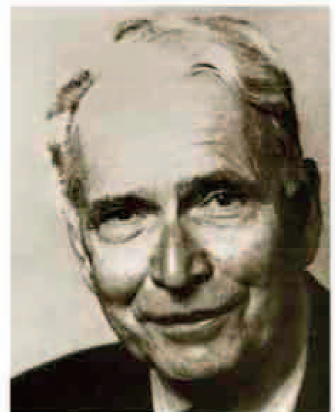
Takahe
Since their rediscovery in 1948, the New Zealand takahe have been moved to predator-free islands as part of a conservation program.



Composition of the Universe
The first elements formed after the Universe formed were the lightest. Cosmologists theorize that the heavier elements were formed inside stars by fusion of atoms.

In the same month, Albert 1 became the **first monkey astronaut** to leave for space onboard a V2 rocket. However, he died of suffocation 39 miles (63 km) into the ascent, before reaching the Karman line that marks the beginning of space at 62 miles (100 km).

Austrian–American chemist **Erwin Chargaff** reported on a study of the **makeup of DNA**—a substance that had recently been shown to be the chemical of heredity. Five years before DNA structure was revealed as a double helix, Chargaff's analysis showed that DNA components called bases occurred in fixed proportions. The proportions of base adenine matched that of thymine, while the proportion of guanine matched cytosine. The double helix model would show that these matches were due to the bases pairing up. Later work revealed that the base sequence along the double helix was the basis for inherited information.



ERWIN CHARGAFF
(1905–2002)

In studying the chemistry of biological molecules, Austrian-born chemist Erwin Chargaff made breakthroughs in a range of topics, such as how blood clots. He moved to the USA when Nazi influence spread through Europe. Chargaff discovered that DNA varied between species, but the proportions of key components were fixed.

On November 20, English ornithologist **Geoffrey Orbell** (1908–2007) made a remarkable discovery when he saw **takahes** in the mountains of South Island, New Zealand. These flightless relatives of rails and moorhens had been thought to be extinct for the previous 50 years.

December Hungarian–British physicist Dennis Gabor patents **hologram technique**

First microwave **oven is sold for commercial use** to a restaurant in Cleveland, Ohio

February 16 Dutch astronomer Gerald Kuiper discovers smallest, innermost **moon of Uranus—Miranda**

March Quantum electrodynamics, **Feynman diagrams** are introduced

April 1 Proportion of elements after the **formation of the Universe** is proposed

April 7 World Health Organization (WHO) established

June 3 Hale telescope is put into operation

June Austrian-American biochemist Erwin Chargaff publishes account of **DNA base composition**

June 11 Albert 1 becomes the **first monkey astronaut**

July 24 WHO holds its first meeting

November 20 Takahe (bird) rediscovered in New Zealand